FACULTY OF INFORMATICS & DESIGN

Diploma in Information & Communication Technology

APPLICATION DEVELOPMENT PRACTICE: ADP470S

SUBJECT GUIDE:	2020
COURSE CODE:	ADP470S
NQF LEVEL:	7
PREREQUISITE SUBJECTS:	PREREQUISISTE SUBJECTS: COURSE CODE: ADP360S OR EQUIVALENT NQF LEVEL: 6

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ORGANISATIONAL COMPONENT

1. INTRODUCTION

Welcome to the Application Development Practice (ADP470S) in Information and Communication Technology in Application Development.

The purpose of this subject is to write efficient program with respect to both time and space parameters using Java as the programming language. You can use NetBeans 8.x as the IDE for your coding. Students will create and work with problems which challenge them to design and implement.

2. GENERAL

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Co-ordinator	Amlan Mukherjee	Engineering 2.27 (A)	(021) 460 3700	mukherjeea@cput.ac.za
Lecturers	Amlan Mukherjee (FT)	Engineering 2.27 (A)	(021) 460-3700	mukherjeea@cput.ac.za
Domain Coordinator	Will be announced later			
Head of Department	B. Kabaso	Engineering 2.30	(021) 460-3713	kabasob@cput.ac.za
Administrative Assistant	D. O'Brien	Engineering 2.49	(021) 460-3685	pillaydl@cput.ac.za
Support Technician	R. Pietersen	Engineering 1.4	(021) 460-4209	pietersenr@cput.ac.za
Support Technician	I. Saliem	Engineering 1.28	(021) 460-3031	saliemi@cput.ac.za
Support Technician	C. Sindayigaya	Engineering 1.28	(021) 460-3031	sindayigayac@cput.ac.za

2.2 Class Times and Venues

As published on http://myclassroom.cput.ac.za and notice boards. The subject consists of one theory periods per week for one and half hrs. The students will be given exercises that they have to do either using their own laptop / pc or during open hrs. in lab.

2.3 Learner Management System (LMS)

https://myclassroom.cput.ac.za/: Blackboard used for general communications and announcements, posting of course material, assignment and announcements. It is the responsibility of the learner to check Blackboard regularly.

3. STUDY MATERIALS

3.1 Prescribed Textbook

Data Structures Using Java:

Moshe J. Augenstein (Author), Yedidyah Langsam (Author), Aaron M. Tenenbaum (Author)

Introduction to Algorithms:

Thomas H. CormenCharles E. Leisersonet

3.2 Recommended Readings

All links provided in BB or during lectures

3.3 Websites

- https://www.youtube.com/playlist?list=PLsyeobzWxl7oRKwDi7wjrANsbhTX0IK0J
- https://www.tutorialspoint.com/video_tutorials/data_structures_and_algorithms/DS-T
- https://w3resource.com/

4. MODULE CREDITS

This module has 0.167 (will be confirmed in Ver. 1.2) credits on NQF level 7. Please refer to the Faculty prospectus on http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus for further information.

4.1 Planning of Time Allocation for Learning Activities

There are 200 notional hours allocated to this module. These notional hours comprise of 60 hours (30%) for contact time and the remainder will be self-study and other tasks, assignments, and tests.

METHOD	% OF YOUR LEARNING TIME
Seminars / Practicals (face-to-face, limited interaction or technologically mediated)	20%
Independent self-study of standard texts and references and specially prepared materials (study guides, books, journals, using online tutorials etc.)	30%
Independent self-study, time needed to prepare for assessment (e.g. tests/exams)	25%
Student learning groups (working with your peers)	25%

5. ETHICAL CONDUCT

5.1 Roles and Responsibilities

Kindly refer to CPUT's Policy Docs and Student Guidelines:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20_Regulations_2018.pdf

5.2 Class attendance and participation

Kindly refer to page 18 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20_Regulations_2018.pdf

5.3 Grievance procedures

Kindly refer to page 61 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20_Regulations_2018.pdf

5.4 Plagiarism

Kindly refer to page 30 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20_Regulations_2018.pdf

**NB: An amended version of the General Handbook will be emailed to you during the first term.*

6. ASSESSMENT

6.1 Assessment policy and regulations

This subject has four assessments. Assessments are a combination of theory and practical tests..

Test queries: Last day for any queries regarding a test is one week after the test was discussed in class. **Thereafter the test mark cannot be changed.**

Absence from a test: If you are absent from a test, a valid medical certificate must be given to your lecturer (via the Secretary) within one week after the test was written or you come to the University. You must also inform your lecturer on the day of the test that you are ill via Email with subject line "Absent for Test".
Due dates must be strictly adhered to. NO late submissions are allowed.
Assignments are administered via Blackboard.

6.2 Assessment Opportunities: Administration

There will be one test per term. There will be a sick test in June for the first semester formal assessments and a sick test in November for the second semester formal assessments.

ASSESSMENT CRITERIA	1ST TERM	2nd TERM	3rd TERM	TOTALS	
Nature of assessment	Practical Test	Written Test	Practical Test	Written Test	100%
Will assessment be included in this term?	Yes	Yes	Yes	Yes	
Total assessment weight for the term	10%	35%	35%	20%	
Will the assessment be moderated?	Yes	Yes	No	Yes	

STUDY COMPONENT

7. SUBJECT SPECIFICATIONS

7.1 Purpose of the Subject

The purpose of this subject is to understand different data structure used in a computer. An IDE like NetBeans can be used to implement the different data structures. The course will also cover different graph algorithms and traversing them. The student also will have the knowledge of different sorting and searching algorithms and compare them for a best solution for a particular case study.

7.2 Subject Structure

7.2.1 Subject Outcomes

SUBJECT OUTCOMES	DESCRIPTION
SO1	Understand the basic knowledge of programming and will be able to write programs using minimum time.
SO2	Knowledge of arrays from 1-d to multidimensional arrays.
SO3	Knowledge of ADT s like stacks, queues and trees. Compare between them for different application
SO4	Implement different sorting and searching algorithms like bubble sort, insertion sort, merge sort, etc. Analyse the Time Complexities of different Algorithms.
SO5	Demonstrate an understanding of graph traversals and hashing. Application of them for real world problems
SO6	Demonstrate an understanding on shortest path algorithms in a graph. Analyse the Time Complexity for discussed Algorithms.

7.2.2 Subject Breakdown Schedule

TERM 1				
SUBJECT OUTCOMES	**THEMES/SPECIFIC OUTCOMES**	**CHAPTER**	**ACADEMIC SCHEDULE**	**ASSESSMENT DATES**
SO1 SO2	THEME 1: RECAP OF JAVA			
	SPO1: Revisit Nested Loops.		03rd Jan –07th Feb	
	SPO2: Introduction to 1-d and 2-d Arrays		10th Feb – 21st Feb	
	SPO3: Bubble Sorting.		24th Feb – 28th Feb	
	SPO4: Recursive Methods and Revision		2nd Mar – 13th Mar	
				Assessment: Practical
TERM 2				
SO3	THEME 2: INTRODUCTION TO ADT			
	SPO1: Recursive Methods Revision		6th Apr – 10th Apr	
	SPO2: Linked-list, Doubly Linked-list, Circular Linked-List		13th Apr – 24^{h} Apr	
	SPO3: Stacks and queues with implementation using both arrays and linked list		27rd Apr –08th May	
	SPO3:Introduction to Graphs and Binary trees with different traversals and Revision		11th May–22nd May	
				Written Test
TEST WEEK				
VACATION				

PUBLIC HOLIDAYS AND OTHER IMPORTANT DATES:

- **21 Mar – Humans Rights Day**
- **10 Apr – Good Friday**
- **13 Apr – Family Day**
- **27 Apr – Freedom Day**
- **01 May – Labour Day**
- **15th June – University Holiday**
- **16 Jun – Youth Day**

TERM 3				
SUBJECT OUTCOMES	**THEMES/SPECIFIC OUTCOMES**	**CHAPTER**	**ACADEMIC SCHEDULE**	**ASSESSMENT DATES**
SO4	THEME 3: Sorting and Searching			
	SPO1: Different Sorting Algorithms and Analyse their Time and Space Complexities		13th Jul – 24th Jul	
	SPO2: Different Searching Algorithms and Analyse their Time and Space Complexities		27nd Jul – 7th Aug	
	SPO3: Compare between the above algorithms with respect to Time and Space		10th Aug – 14rd Aug	
				Practical Test
TEST/PROJECT/SICK TEST WEEK				
RECESS				

TERM 4				
SUBJECT OUTCOMES	**THEMES/SPECIFIC OUTCOMES**	**CHAPTER**	**ACADEMIC SCHEDULE**	**ASSESSMENT DATES**
SO5 SO6	THEME 4: Advanced DS topics			
	SPO1: Hash-table, hashing with open addressing, hashing with separate chaining and their application in real world		7th Sep – 11th Sep	
	SPO2: Graph traversals, depth first traversal. Breadth first traversal with the complexities involved		16th Sep– 25thSep	
	SPO3: Introduction to B Trees, B+ Trees, etc.		28th Sep – 2nd Oct	
	SPO3: Minimum spanning trees (MST) and shortest path algorithms in graph		5th Oct – 16th Oct	
				Written Test
TEST/PROJECT/SICK TEST WEEK				
VACATION				

PUBLIC HOLIDAYS AND OTHER IMPORTANT DATES:

- **09 Aug– National Women's Day**
- **10Aug – Women's Day Observed**
- **24 Sep – Heritage Day**
- **25 Sep – University Holiday**
- **25 Dec – Christmas Day**
- **26 Dec – Day of Good Will**

7.3 Articulation with other subjects in the programme

This subject is linked to concepts to the programming knowledge taught in Diploma Course.

7.4 Learning presumed to be in place

Knowledge of any programming language is the pre-requisite for this subject.

7.5 Critical Cross-field Outcomes

CRITICAL CROSS-FIELD OUTCOMES	DESCRIPTION
CCFO1	Identify and solve problems with responses which display responsible decisions, using critical thinking, have been made.
CCFO2	Work effectively with others as a member of a team, group, organisation or community.
CCFO3	Organise and manage oneself and one's activities responsibly and effectively.
CCFO4	Collect, analyse, organise and critically evaluate information.
CCFO5	Communicate effectively, using visual, mathematical, and/or language skills in the modes of written and/or oral presentation.
CCFO6	Use science and technology effectively and critically, showing responsibility towards the environment and the health of others.
CCFO7	Demonstrate an understanding of the world as a set of related systems by recognising that problem-solving contexts do not exist in isolation.
CCFO8	Contribute to the full personal development of each learner and the social and economic development of the society at large, by making it the underlying intention of any programme of learning to create awareness in individuals about the importance of: i) reflecting on and exploring a variety of strategies to learn more effectively; ii) participating as responsible citizens in the life of local, national and global communities; and iii) being culturally and aesthetically sensitive across a range of social contexts. (RSASAQA, 2001:24)

8. STUDY UNITS/STUDY THEMES

8.1 Study Units

Refer to 7.2.2 Subject Breakdown Structure.

8.2 Assessment Opportunities and Assessment strategy

Refer to 6.2 Assessment Opportunities: Administration for the linkages.

8.3 Subject Outcomes and Associated Assessment Criteria

Refer to 7.2.2 Subject Breakdown Structure for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment and project work.

8.4 Self-study Activities

Instructions will be given during lecture hrs. or via black board.

9. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY

STUDENT ACKNOWLEDGEMENT

2019

DIPICTA: INFORMATION TECHNOLOGY

APPLICATION DEVELOPMENT FUNDAMENTALS II

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **APPLICATION DEVELOPMENT FUNDAMENTALS II** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname: …………………………………………………….
(Print)

Student Number: ……………………………….

Contact number: ……………………… (home/res) Contact number: ……………………… (cell)

Signature: ……………………… Date: ………………………